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TITLE: ResoTrack – A Resource Management System

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ABSTRACT

Efficient management of organizational resources and inventory is essential for maintaining productivity, reducing delays, and ensuring smooth day-to-day operations. Many institutions and businesses still rely on manual records or disconnected systems, which often lead to inaccurate stock data, poor tracking, approval delays, and inefficient allocation of resources.

To address these challenges, we developed **ResoTrack: Smart Resource & Inventory Management System**, a web-based application designed to digitize and streamline the complete lifecycle of resource management. The system enables organizations to manage inventory, monitor stock levels, process resource requests, and control approvals through a secure role-based environment.

ResoTrack is developed using the **MERN Stack (MongoDB, Express.js, React.js, Node.js)**, providing a modern, scalable, and responsive full-stack solution. The platform supports three primary user roles: **Admin**, **Manager**, and **Staff**. Admins can manage users and oversee system-wide operations, Managers can supervise inventory and approve requests, while Staff members can browse available resources and submit requests.

Key features of the system include inventory tracking, request approval workflow, notification system, low-stock alerts, analytics dashboard, and secure authentication using JWT-based access control.

The proposed system improves transparency, minimizes manual effort, enhances accountability, and ensures efficient utilization of organizational resources. ResoTrack demonstrates how modern web technologies can be used to solve real-world inventory and resource management challenges effectively.

INTRODUCTION

In today's fast-paced organizational environment, efficient handling of resources and inventory is crucial for smooth operations and improved productivity. Resources such as hardware equipment, software licenses, office assets, and operational materials must be properly tracked, allocated, and maintained. Traditional methods of managing such resources through manual registers, spreadsheets, or disconnected systems often result in errors, data redundancy, delays, and lack of accountability.

With the increasing need for automation and real-time data access, organizations require a centralized digital platform that can simplify inventory handling and resource allocation. A smart management system can help reduce manual workload, improve transparency, and support better decision-making through accurate records and analytics.

To solve these problems, we developed **ResoTrack: Smart Resource & Inventory Management System**. ResoTrack is a web-based platform that allows organizations to manage inventory efficiently, track stock availability, process resource requests, and control approvals through role-based access. The system ensures that each user can perform only authorized actions according to their designated role.

The project is developed using the **MERN Stack**, consisting of **MongoDB** for database management, **Express.js** and **Node.js** for backend services, and **React.js** for the frontend user interface. This combination provides scalability, responsiveness, and efficient performance.

ResoTrack is designed to modernize resource management processes by replacing outdated manual systems with a secure, user-friendly, and automated solution.

PROBLEM BACKGROUND

Many organizations, institutions, and warehouses still depend on manual methods such as spreadsheets, paper records, or scattered digital files to manage resources and inventory. These traditional approaches are time-consuming, prone to human error, and difficult to maintain as the volume of resources increases.

In the absence of a centralized system, organizations often face problems such as inaccurate stock records, duplicate entries, misplaced assets, delayed approvals, and poor visibility of available resources. It also becomes difficult to identify low-stock items in time, resulting in operational delays and inefficient planning.

Another major challenge is the lack of accountability and secure access control. When multiple users handle the same data without proper permissions, unauthorized changes and data inconsistency can occur. Managers may also struggle to monitor requests, approvals, and overall resource movement efficiently.

Therefore, there is a need for a smart, centralized, and role-based digital platform that can automate inventory operations, simplify request handling, provide timely notifications, and improve overall resource utilization.

ResoTrack is proposed as a solution to address these challenges by offering a modern web-based Resource & Inventory Management System that improves efficiency, transparency, and control.

OBJECTIVES

The main objective of **ResoTrack** is to develop a smart and centralized system for efficient management of organizational resources and inventory. The system is designed to replace manual processes with a secure and automated digital platform.

The specific objectives of the project are as follows:

1. **To centralize inventory management** by maintaining all resource records in a single database system.
2. **To provide role-based access control** so that Admins, Managers, and Staff users can perform only authorized operations.
3. **To simplify resource request handling** by allowing staff members to request available resources through an organized workflow.
4. **To streamline approval processes** by enabling managers or admins to approve, reject, or manage pending requests.
5. **To monitor stock levels effectively** and generate alerts when resources reach low quantity levels.
6. **To improve data accuracy and transparency** by reducing duplicate entries and maintaining updated records.
7. **To provide analytical insights** through dashboards and summaries for better planning and decision-making.
8. **To build a scalable web application** using the MERN stack that can be enhanced with future features such as barcode scanning, multi-warehouse support, and real-time tracking.

TECHNOLOGY STACK

The development of **ResoTrack** is based on the MERN Stack, which provides a complete full-stack JavaScript solution for building scalable and responsive web applications. The technologies used in the project are as follows:

Frontend Technologies

1. **React.js**
Used for building a dynamic and interactive user interface through reusable components.
2. **Vite**
Used as the frontend build tool for faster development, optimized builds, and improved performance.
3. **Redux Toolkit**
Used for centralized state management to handle authentication, inventory data, requests, and user-related operations.
4. **CSS / Tailwind CSS** (*if applicable*)
Used for styling, responsive layouts, and improving the visual appearance of the application.

Backend Technologies

1. **Node.js**
Provides the runtime environment for executing JavaScript on the server side.
2. **Express.js**
Used to build RESTful APIs, manage routes, middleware, and backend logic.
3. **JWT (JSON Web Token)**
Used for secure user authentication and session management.
4. **bcryptjs**
Used for password hashing and securing user credentials.

Database

1. **MongoDB**
Used as the NoSQL database to store users, resources, requests, and notification data.
2. **Mongoose**
Used as an Object Data Modeling (ODM) library for MongoDB schema design and database operations.

Additional Tools & Services

1. **Nodemailer**
Used for sending low-stock email alerts and notifications.
2. **Git & GitHub**
Used for version control, collaboration, and source code management.

3. **Postman**

Used for API testing and backend debugging during development.

The selected technology stack ensures flexibility, scalability, maintainability, and efficient performance for the ResoTrack system.

SYSTEM ARCHITECTURE

ResoTrack follows a three-tier web application architecture consisting of the Presentation Layer, Application Layer, and Data Layer. This structure helps in maintaining scalability, modularity, and easier management of the system.

1. Presentation Layer (Frontend)

The frontend of the system is developed using **React.js** with **Vite** as the build tool. It provides a responsive and interactive user interface for different users such as Admin, Manager, and Staff.

Main frontend responsibilities include:

- User login and registration interfaces
- Dashboards for different user roles
- Resource catalog display
- Request submission forms
- Notifications and alerts
- Data visualization and summaries

2. Application Layer (Backend)

The backend is developed using **Node.js** and **Express.js**, which handle all server-side logic and communication between the frontend and database.

Main backend responsibilities include:

- User authentication and authorization
- Role-based access control
- Inventory CRUD operations
- Request approval and status updates
- Notification generation
- Low-stock email alert handling
- REST API management

3. Data Layer (Database)

The system uses **MongoDB** as the primary database with **Mongoose** for schema management and data interaction.

The database stores:

- User accounts and roles
- Resource inventory records
- Request details and statuses

- Notification records
- System activity data

Working Flow of the System

1. User logs into the system using secure credentials.
2. Based on the assigned role, the user is redirected to the relevant dashboard.
3. Staff users browse resources and submit requests.
4. Managers/Admins review pending requests and take actions.
5. Inventory updates automatically when approved actions are processed.
6. Notifications and alerts are generated for important events such as approvals or low stock.
7. Dashboards display updated summaries and operational insights.

Advantages of the Architecture

- Clear separation of concerns
- Easy maintenance and future upgrades
- Better scalability for larger organizations
- Secure access control through backend APIs
- Efficient communication between frontend and database

FUNCTIONAL MODULES

ResoTrack is divided into multiple functional modules to ensure smooth management of resources, users, and operational workflows. Each module is designed to handle a specific part of the system efficiently.

1. Authentication Module

This module manages secure user access to the system.

Functions:

- User login and registration
- Password encryption using bcryptjs
- JWT-based authentication
- Session validation
- Role-based route protection

2. User Management Module

This module is mainly controlled by the Admin user.

Functions:

- Create new users
- Assign roles such as Admin, Manager, and Staff
- View existing users
- Manage user accounts

3. Resource Management Module

This module handles all inventory and resource-related operations.

Functions:

- Add new resources
- Update resource details
- Delete obsolete resources
- Manage quantities and categories
- Monitor stock availability

4. Request Management Module

This module allows staff users to request resources.

Functions:

- Browse available resources
- Submit requests
- Add notes if required
- View request history
- Track request status

5. Approval Module

This module is used by Managers and Admins.

Functions:

- View pending requests
- Approve requests
- Reject requests
- Finalize request lifecycle
- Prevent unnecessary updates after final approval

6. Notification Module

This module improves communication inside the system.

Functions:

- Request approval notifications
- Rejection alerts
- Low-stock notifications
- Email alerts for critical stock levels

7. Dashboard & Analytics Module

This module provides operational summaries and insights.

Functions:

- Total resources count
- Request statistics
- User summaries
- Low-stock indicators
- Visual charts and analytics

These modules together make ResoTrack a complete and practical Resource & Inventory Management solution.

DATABASE DESIGN

The database for **ResoTrack** is developed using **MongoDB**, a NoSQL document-oriented database. MongoDB stores data in flexible JSON-like documents, making it suitable for dynamic and scalable web applications. **Mongoose** is used as the Object Data Modeling (ODM) library to define schemas and manage database operations.

The major collections used in the system are as follows:

1. Users Collection

This collection stores information related to all registered users of the system.

Fields may include:

- Name
- Email
- Password (hashed)
- Role (Admin / Manager / Staff)
- Created Date

Purpose:

Used for authentication, authorization, and user management.

2. Resources / Inventory Collection

This collection stores all available organizational resources and stock items.

Fields may include:

- Resource Name
- Category
- Quantity
- Description
- Low Stock Threshold
- Status
- Created Date

Purpose:

Used to track inventory levels, manage stock, and monitor availability.

3. Requests Collection

This collection stores requests submitted by staff users for required resources.

Fields may include:

- Requester ID
- Resource ID / Requested Items
- Request Note
- Status (Pending / Approved / Rejected / Cancelled)
- Request Date
- Updated Date

Purpose:

Used to manage request workflow and approval processes.

4. Notifications Collection *(if implemented)*

This collection stores system-generated notifications.

Fields may include:

- User ID
- Notification Message
- Type
- Read Status
- Timestamp

Purpose:

Used for approval alerts, stock warnings, and operational updates.

Relationships Between Collections

- One user can create multiple requests.
- One request can contain one or more resources.
- Notifications can be linked to users or system events.
- Resources are updated based on approved requests.

Benefits of MongoDB for This Project

- Flexible schema design
- Easy scalability
- Fast JSON-based data storage
- Smooth integration with Node.js and React
- Efficient handling of dynamic application data

KEY FEATURES

ResoTrack offers a range of practical features designed to simplify resource handling, improve inventory visibility, and streamline organizational workflows.

1. Secure Authentication System

- User login and registration with encrypted passwords
- JWT-based secure session management
- Protected routes based on user roles

2. Role-Based Access Control

Different dashboards and permissions for:

- **Admin** – Full system control and user management
- **Manager** – Inventory supervision and request approvals
- **Staff** – Resource browsing and request submission

3. Resource Catalog Management

- Add new resources to inventory
- Edit resource details
- Delete obsolete items
- Categorize resources efficiently
- Track available quantities

4. Request & Approval Workflow

- Staff users can request required resources
- Requests move through Pending / Approved / Rejected stages
- Managers/Admins can review and take actions
- Finalized requests are protected from unnecessary updates

5. Low Stock Monitoring

- Detects items reaching critical stock levels
- Sends automated alerts for timely restocking

6. Notification System

- Approval and rejection updates
- Stock-related alerts
- Activity notifications within the system

7. Dashboard & Analytics

- Total users count
- Total requests count
- Inventory summaries
- Low-stock indicators
- Operational charts and statistics

8. Responsive User Interface

- Clean and user-friendly design
- Works across desktop and standard browser environments
- Easy navigation for different user roles

9. Scalable Architecture

- Built on MERN stack for future expansion
- Can be extended with barcode scanning, multi-location inventory, and advanced reporting

These features make ResoTrack a practical and modern solution for resource and inventory management.

SCREENSHOTS

Fig. 1 – Login Page

Displays secure login interface for Admin, Manager, and Staff users.

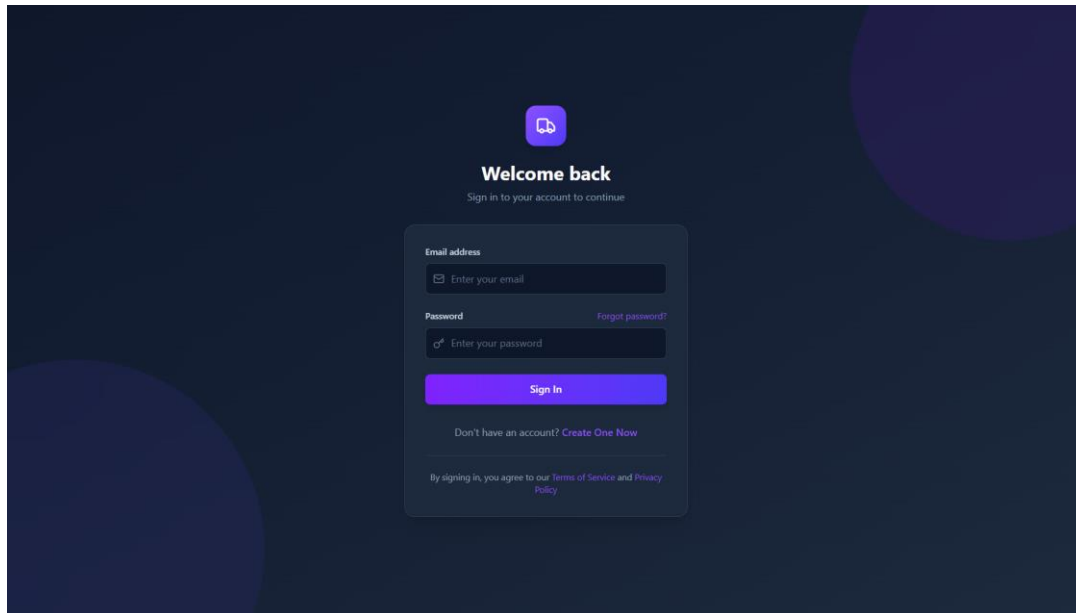


Fig. 2 – Admin Dashboard

Shows system analytics such as total users, total requests, low-stock alerts, and summaries.

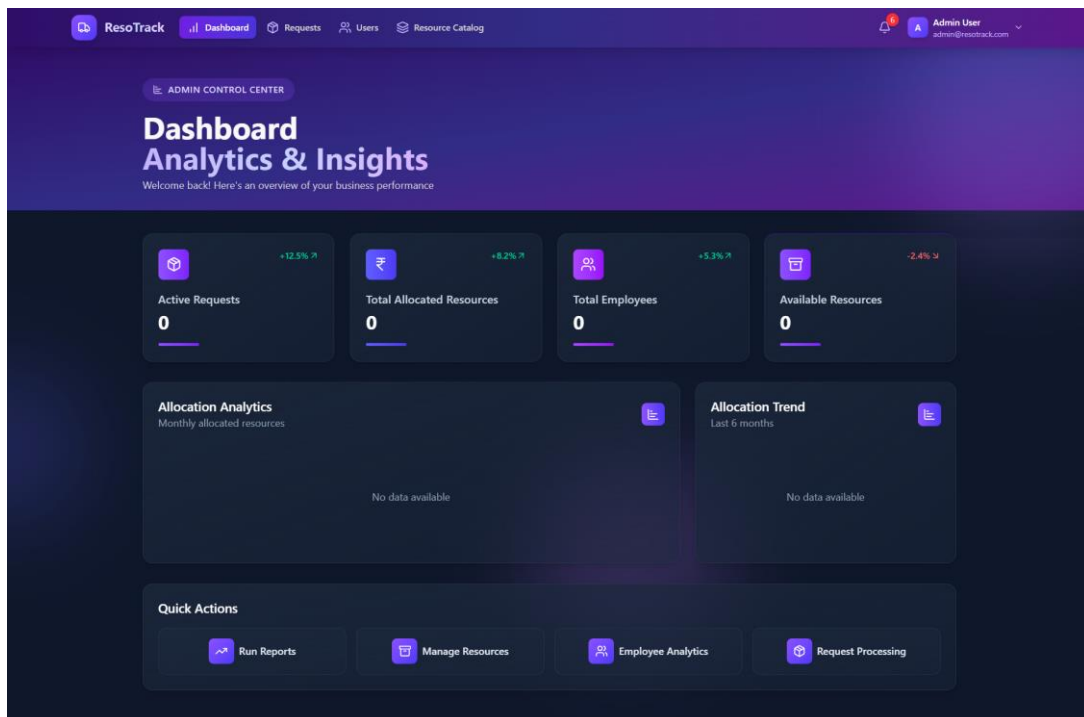


Fig. 3 – Resource Catalog

Displays available resources with categories, stock quantity, and management options.

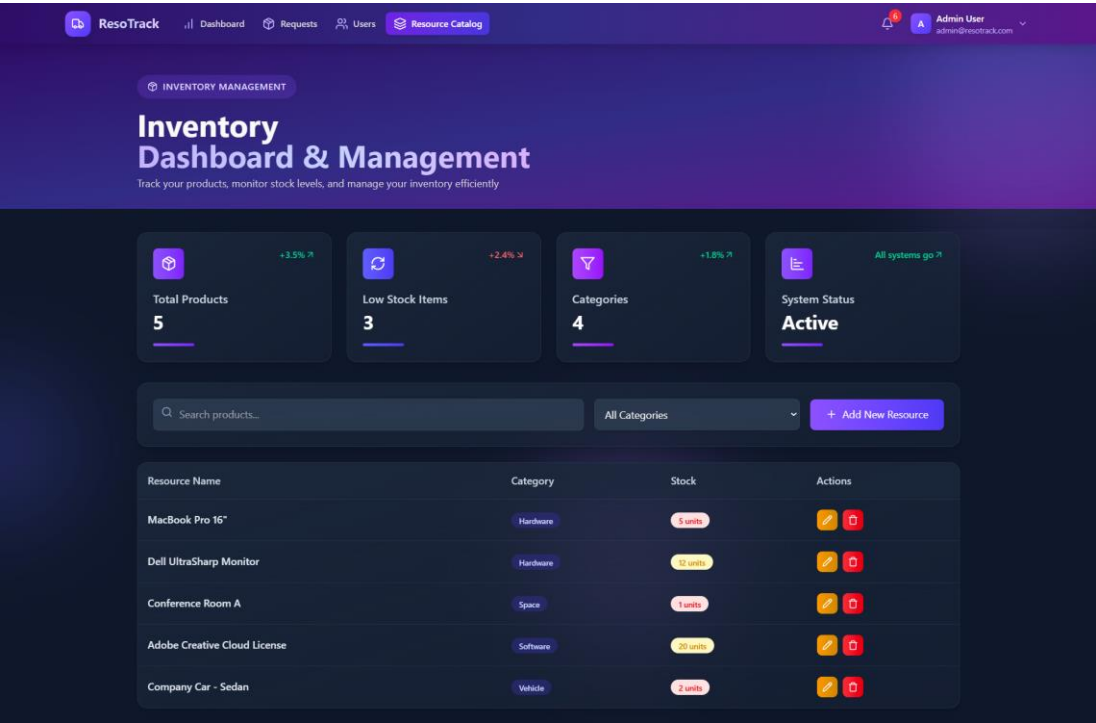


Fig. 4 – Request Submission Page

Shows staff interface for selecting resources and submitting requests.

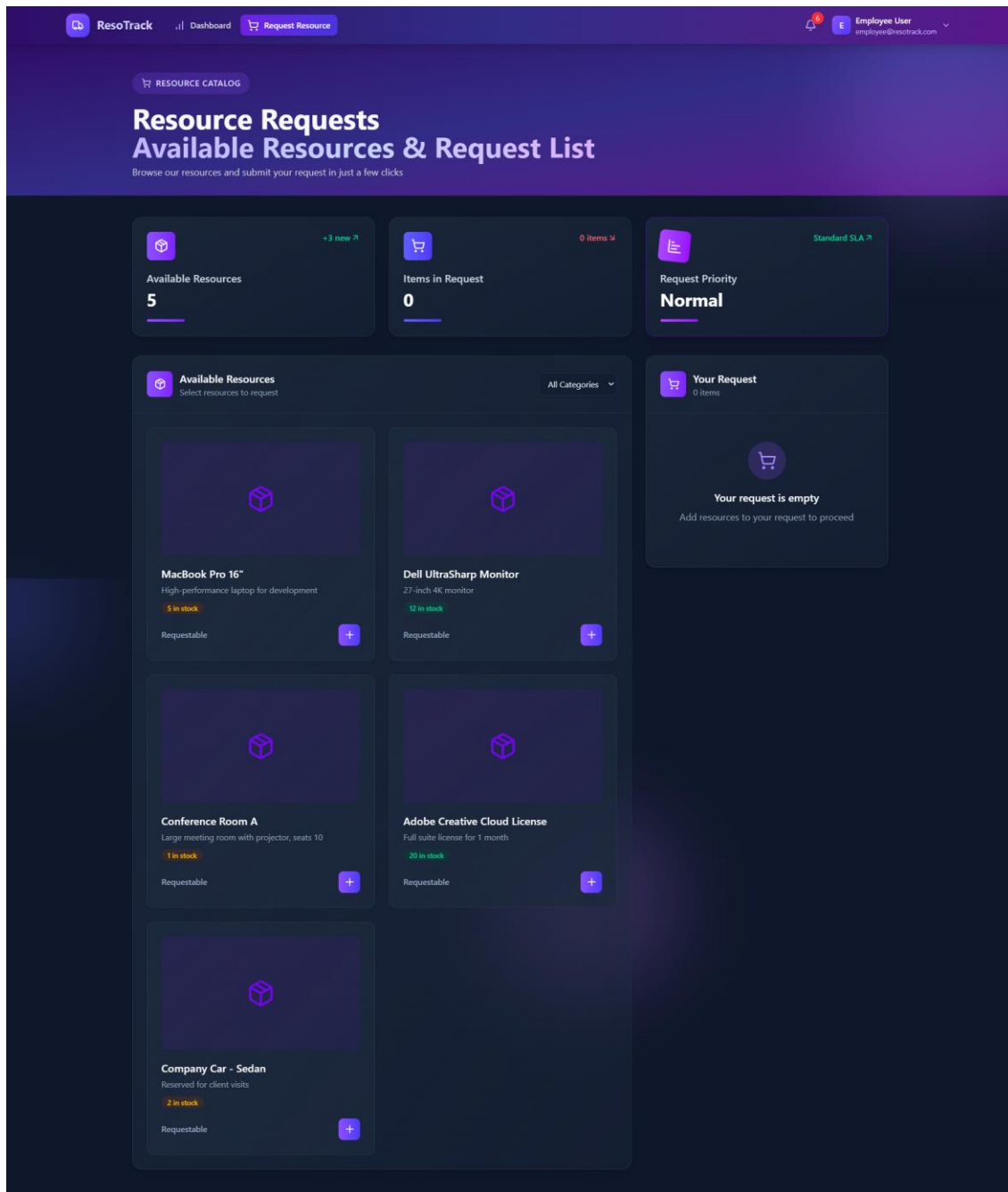


Fig. 5 – Approval Management Panel

Shows Manager/Admin interface for reviewing pending requests and approving or rejecting them.

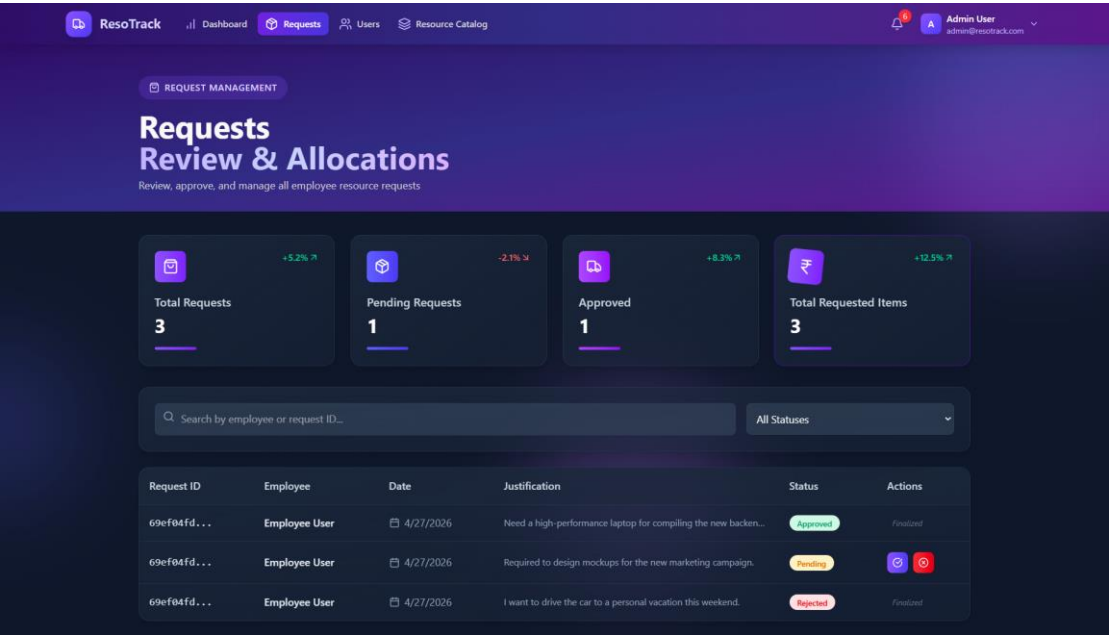


Fig. 6 – User Management Module

Displays Admin interface for creating and managing users.

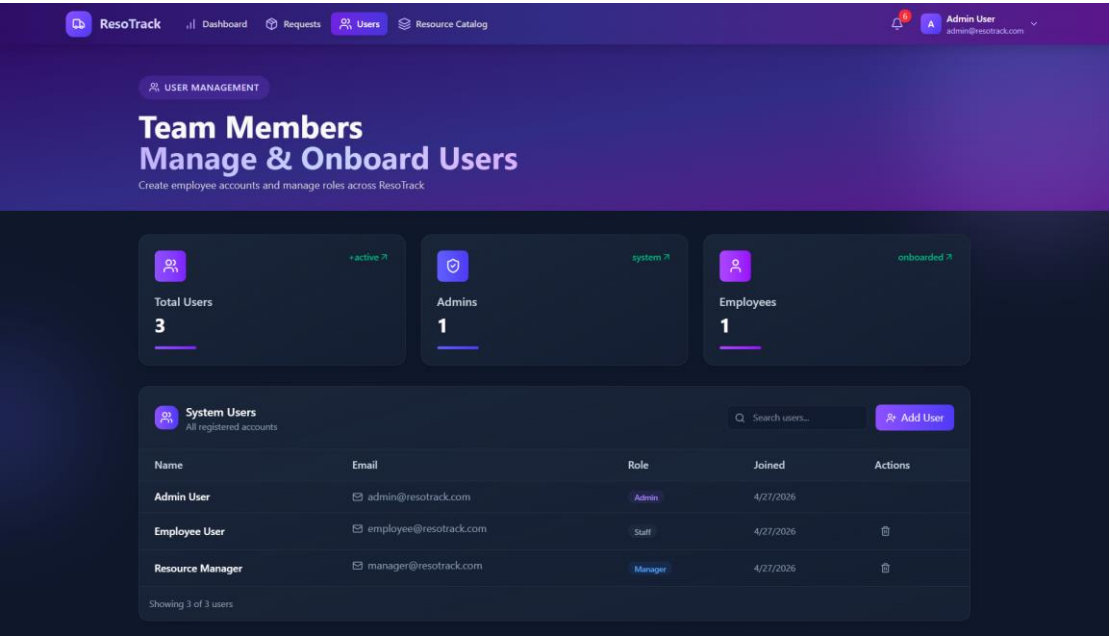


Fig. 7 – Notifications Panel

Shows alerts related to approvals, rejections, and low stock events.

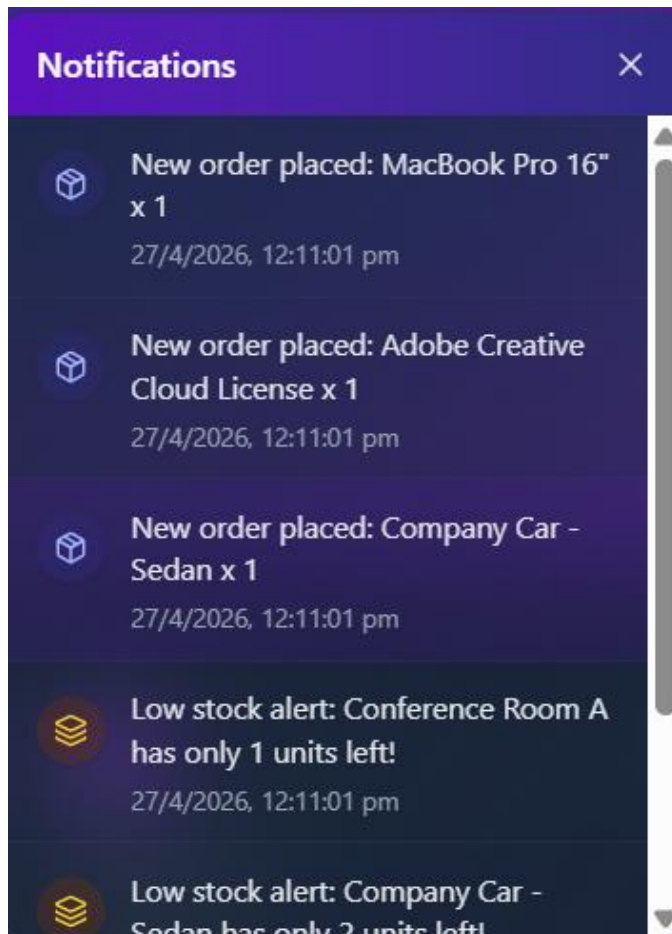
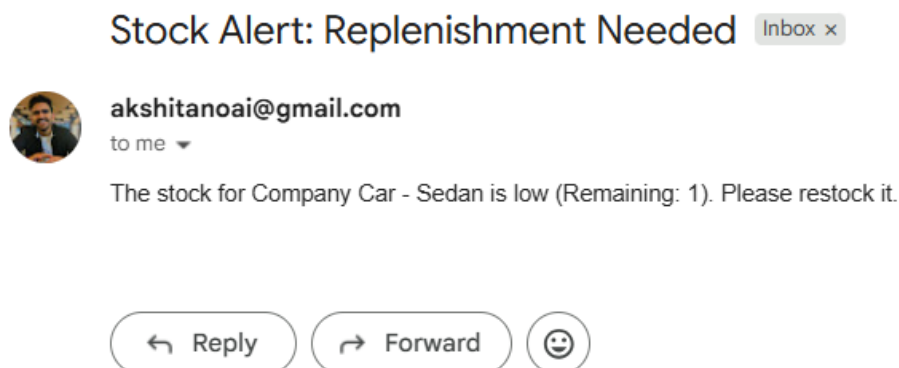


Fig. 8 – Low Stock Email Alert

Displays sample email notification received when stock reaches critical threshold.



FUTURE SCOPE

Although ResoTrack currently provides the essential features required for smart resource and inventory management, the system can be further enhanced in the future with advanced capabilities to improve scalability, automation, and user experience.

1. Barcode / QR Code Integration

Resources can be tagged with barcode or QR codes for faster identification, check-in, check-out, and stock verification.

2. Multi-Warehouse Support

The system can be extended to manage inventory across multiple warehouse locations or branches from a centralized dashboard.

3. Real-Time Updates

Live synchronization using technologies such as WebSockets can provide instant updates for requests, stock changes, and notifications.

4. Advanced Analytics & Forecasting

AI/ML-based analytics can be added to predict stock demand, optimize procurement planning, and analyze usage patterns.

5. Mobile Application

A dedicated Android/iOS mobile application can provide on-the-go access for managers and staff.

6. Audit Logs & Activity Tracking

Detailed logs can be maintained for every action performed in the system to improve accountability and security.

7. Email / OTP Based Onboarding

Automated onboarding, password reset, and verification workflows can be added for better user management.

8. Department-Based Allocation

Resources can be assigned and tracked department-wise for more organized management.

9. Cloud Deployment

The system can be deployed on cloud platforms for wider accessibility, backups, and scalability.

With these enhancements, ResoTrack can evolve into a more intelligent, enterprise-ready, and large-scale inventory management solution.

CONCLUSIONS

ResoTrack successfully demonstrates the development of a modern web-based **Resource & Inventory Management System** that addresses common challenges associated with manual inventory handling and unstructured resource allocation processes.

The project provides a centralized platform where resources can be managed efficiently, requests can be processed through a structured approval workflow, and users can operate according to their assigned roles through secure access control. By integrating inventory tracking, notifications, low-stock alerts, and dashboard insights, the system improves visibility, accountability, and operational efficiency.

The use of the **MERN Stack** enables the application to be scalable, responsive, and maintainable while offering a smooth user experience. The project also reflects the practical implementation of concepts such as full-stack development, database design, authentication, API integration, and role-based authorization.

Overall, ResoTrack is a useful and practical solution for organizations seeking to modernize their inventory and resource operations. With future enhancements, the system has the potential to grow into a more advanced enterprise-level platform.

REFERENCES & PROJECT REPOSITORY

1. React.js Official Documentation – <https://react.dev>
2. Node.js Official Documentation – <https://nodejs.org>
3. MongoDB Documentation – <https://www.mongodb.com/docs>
4. Project Source Code Repository: <https://github.com/AkshitG0914/ResoTrack>